Design decisions put to the owner

Every decision here was a fork the evidence could not settle on its own: either it changed the architecture, or it traded cost against reliability. Each entry gives the question as it was asked, the options offered, what was chosen, and why the question existed at all.

Dates are 2026. The standing rule from 22 September — **safety and reliability before cost, size and convenience** — is what several of these answers turn on.

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1. The harness connector

Asked: the specification (§7, deviation 1) called for the OEM 94-way connector to be replaced by functionally-split sealed connectors plus an adapter harness. The physical part had since been identified from photographs of the machine. Keep the deviation, or put the OEM connector on the board?

Chosen: the OEM Bosch 94-way on the board, no adapter. Spec deviation 1 is recorded as reversed.

Consequence: the board-side land pattern is now load-bearing and has to come from a Bosch distributor. Neither part number returns a public catalogue hit.

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2. Four part-selection findings, approved together

These four were presented as recommendations and approved with one "ok" on 22 September. Each one came out of choosing a part and discovering the drawn circuit could not accept it.

2a. Where the injector and EGR current comes from — POWER-PATH

The problem: every high-current load (injector hold ≈10 A, EGR up to 6.4 A stall, relay coils) draws through connector pin 21, a 20 A fuse, the EMI filter and a reverse-battery FET. The reverse-battery block sized that FET for **5 A**, and the EMI filter block never stated a DC current at all. A 22 µH inductor that does not saturate at 15–20 A is a large part. Meanwhile pins 04 and 06 are separate battery feeds, fused 10 A each in the loom, that this board uses only to measure voltage.

Options: move the injector and EGR supply onto pins 04/06 with their own protection, leaving pin 21 for the electronics; or keep everything on pin 21 and resize the fuse, filter and reverse FET for the full load,

with one contact carrying it.

Chosen: move the high-current loads to pins 04/06. Status: built 8 October 2026 (`power_path.cir`, power_input sheet).

How it was built took three more questions, recorded in §5.

2b. Boost converter stability while cranking — BOOST-SLOPE

The problem: the injector boost controller (TPS40210-Q1) adds a fixed internal slope-compensation ramp. Its Equation 9 caps the current-sense resistor at `VDD·L·fSW / (60·(VOUT + VD - VIN))`, and 80% of that is the recommended value. The drawn 82 mΩ passes at 13.5 V (ceiling 99.7 mΩ) and **fails at the 6 V cranking corner** (ceiling 44.9 mΩ, so 1.83× over): the converter goes subharmonic exactly when the engine is being started. Worse, a 330 µH inductor that saturates above the 2.2 A current limit is not a part that exists in surface mount — the search found 330 µH parts rated 0.2–0.4 A — and the buildable direction, a smaller inductor, lowers the ceiling further (100 µH allows 30 mΩ).

Options: feed the controller's supply pin from the output side, which is TI's own remedy and steepens the ramp, and use 100 µH; or accept subharmonic operation while cranking.

Chosen: feed VDD from the clamped output side, 100 µH. Revised 8 October: 150 µH (Würth 7447709151), see §8. Status: built 8 October 2026.

2c. Sensor-supply protection — SENSOR-PTC

The problem: the requirement asked for a resettable fuse of "R25 = 3.0 Ω nominal, −30% spread". Real PTCs are not specified that way: Bourns MF-USMF020 gives a **range, 0.40–5.00 Ω**, and at its 0.40 Ω minimum it contributes 14% of the series resistance the fault-current ceiling assumed. Its hold current also halves from 0.20 A at 23 °C to 0.10 A at 85 °C, against a ≥200 mA requirement.

Chosen (first pass): a fixed series resistor sets the fault ceiling and the fuse only trips. This was later superseded — see decision 4.

2d. The 3.0 V and 3.3 V input clamps — LOWV-CLAMP

The problem: fifteen zeners clamp the analog, speed and discrete inputs. They were specified at 5 mA but sit at tens of microamps, where a sub-5 V zener's knee is soft. The datasheet allows **10 µA of leakage at only 1 V.** On a discrete input, 10 µA through the ~28 kΩ source is 0.28 V against a logic-high margin of 0.36–0.41 V while cranking; on an analog input it is a signal-dependent error on the ADC. The simulation's idealised sharp-knee zener hid all of it.

Chosen: replace them with a low-leakage clamp to the 3.3 V rail (BAV199). Status: done for the analog and cam inputs; the discrete inputs needed a different answer — see decision 3.

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3. The discrete switch inputs

Why it came back: BAV199 works where a clamp only conducts on a fault. The five switch inputs clamp **continuously**, because a 6–40 V input cannot be divided into logic range otherwise: reading a logic high at 6 V needs a divider ratio above 0.385, and staying under 3.3 V at 40 V needs one below 0.0825. Those do not overlap. Three candidate clamps were checked and all three fail:

Candidate	Why it fails
3.0 V zener (BZX84-C3V0-Q)	soft knee at µA; the leakage that started this
Rail clamp (BAV199-Q to 3V3)	parks the pad at 3.9 V continuously — beyond the pin's rating, as an operating point rather than a fault
Shunt reference (LM4040-Q1 3.0 V)	draws ~35 µA of bias below breakdown (TI SNOS633N, Fig. 5-4); the cranking-dip input can supply only 20 µA, so the node sags to ~2.6 V with no guaranteed bound, and it fails outright on the pull-up channels

Options offered: an NPN buffer per input; a stronger drive plus the LM4040, with special cases for the two contact-to-ground channels; or a dedicated automotive switch-detection IC on SPI.

Chosen: the NPN buffer. 10 kΩ into the base, 4.7 kΩ and 4.7 µF to ground, a BAV199-Q diode holding the base against reverse battery, and the transistor's collector as the MCU pin with a 10 kΩ pull-up to 3.3 V. The pad now only ever sees 0–3.3 V, and nothing on it has to conduct continuously.

What the safety rule changed inside this decision: the input resistor was first drawn at 100 kΩ, which draws 0.11 mA through a closed contact at 12 V. Non-gold switch and relay contacts at that current oxidise into an intermittent open — a field failure, not a bench one. 10 kΩ gives 1.1 mA. Everything is checked at the worst corner rather than typical: hFE 30 (datasheet minimum 60, halved for −40 °C) still pulls the pad to 0.03 V from a 6 V input, and a typical part at 125 °C stays off below a 1.3 V input.

Cost: the logic is inverted — pad low means input active — which is recorded in the pin map, and the failure modes (transistor open reads inactive, collector-emitter short reads active) are handed to the firmware plausibility spec, because one polarity cannot make every channel fail safe.

Status: done.

4. The 5 V sensor supply

Why it came back: the fix approved in 2c does not survive the real source impedance. The sensor rail block modelled the 5 V supply as ideal behind 0.1 Ω, but it comes from a **1 A buck**. So:

- a shorted sensor harness hits the buck's current limit long before any fuse trips; 5V_MAIN collapses, 3V3_MCU follows it, and the supervisor shuts the engine down. One chafed wire stops the machine.
- a sensor supply wire shorted to **battery** back-feeds 24 V into 5V_MAIN through the fuse. No series resistor or PTC prevents that.

Options offered:

Option	What it buys	What it costs
Tracker per group + overvoltage cut-off	each group survives a short to ground *and* to battery; other groups and the MCU are unaffected	about 8 more parts
Series resistor + PTC (as approved in 2c)	cheapest	a harness short browns out the MCU; a battery short back-feeds the rail
Trackers fed directly, with a lower main TVS	fewer parts	rewrites the whole board's transient design, and needs the alternator question (U16) answered first

Chosen: a tracking regulator per sensor group (TPS7B4253-Q1, whose output survives −1 to +45 V and limits at 301–520 mA, tracking 5V_MAIN to ±4 mV) behind a 100 V-class overvoltage cut-off.

Why the cut-off is needed: the tracker's input pin is rated 45 V and this board's battery rail reaches 73.3 V for 50 µs on ISO 7637-2 pulse 2a. The cheap alternatives were worked through and discarded: a series resistor with a TVS must clamp below 45 V, so it would conduct through the entire 400 ms load dump (about 7 J), and any series resistance breaks the 0.52 V of dropout headroom at a 6 V cranking input; an emitter or source follower loses its own Vbe/Vgs at that same corner. A cut-off instead disconnects for the 50 µs pulse, while a 47 µF capacitor holds the sensors up (0.16 V of droop), and stays connected through the 40 V dump.

Status: built 8 October 2026. The cut-off controller is the **TPS48110-Q1** (see §7). `sensor_rail.cir` was rewritten around it.

5. How the POWER-PATH feeds are built (8 October)

§2a decided that the loads move to pins 04/06. These three questions decided how. Each was asked with a recommended option, and each was answered with it.

5a. Which loads go on which pin

Asked: how should the high-current loads be split across pins 04 and 06?

Options: - **04 injector, 06 EGR** (recommended). Pin 04 feeds the injector hold switches and the boost converter input; pin 06 feeds the EGR H-bridge. Separate rails, separate protection: an EGR short blows only its own fuse, and no two fuses are paralleled. The injector averages well under its fuse rating (10 A only during holds of about 1–3 ms), and the EGR's 6.4 A stall is inside its 10 A fuse. - **04 + 06 paralleled** onto one high-current rail. Two fuses in parallel never share evenly: one blows first and dumps the whole load on the other.

Chosen: 04 injector (`VBAT_INJ`), 06 EGR (`VBAT_EGR`).

Why it was a question: U7, whether the OEM uses pins 04/06 as load feeds at all, is still open. What makes them usable is the harness itself: each pin has its own 10 A loom fuse, and a fuse is sized to protect its wire.

5b. Reverse-battery protection on the new rails

Asked: what should provide reverse-battery protection on the two new rails?

Options: - **A P-FET per rail** (recommended). Same passive topology as Q1 on pin 21: a gate resistor to ground plus a zener, and no controller IC, so nothing has to be rated for the +73 / −48 V swing. It needs a lower-RDS(on) part, chosen from a retrieved datasheet, for 6.4 A continuous at a 105 °C board. The reversed-battery fault is removed rather than managed. - **Fuse only.** No series element: a reversed battery drives current through the bridge FETs' body diodes until the 10 A loom fuse blows. No drop or heat in normal running, but it relies on the FETs surviving the fuse's clearing time — a managed failure, which the standing rule argues against.

Chosen: a P-FET per rail — SQM120P10-10m1L (Vishay doc 76943, retrieved 8 October).

Why it was a question: pin 21's SQD50P08-25L could not simply be copied. At the EGR's 6.4 A stall it would dissipate 1.3–1.8 W, which takes a DPAK on a 105 °C board past its junction limit. The chosen part runs at 139 °C (155 °C while cranking), checked in `power_path.cir`.

5c. What stays on pin 21

Asked: which loads stay on pin 21, the filtered electronics feed?

Options: - **Buck, relays, metering** (recommended). Pin 21 keeps the buck (all the electronics), the relay coils (~84 mA) and the metering valve's freewheel diode (0.86 A max; the valve itself is powered by the

OEM's 20 A fuse 8F1). The boost controller's VDD stays on VBAT_PROT until BOOST-SLOPE moves it. Pin 21 then carries a couple of amps. - **Buck only.** Move the relay coils and the metering freewheel onto the new rails as well. Pin 21 becomes electronics-only, but the relays and the metering valve then depend on the 04/06 fuses too.

Chosen: buck, relays and metering stay on pin 21.

Consequence: L101, L102 and F101 are unblocked and are now sized for pin 21's reduced load, which still has to be stated as a number.

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6. The toolchain on a new machine (7–8 October)

Setting the project up on macOS raised three questions. They are about tools, not the circuit, but two of them decide which simulator and which libraries the circuits are checked against.

6a. Which KiCad

Asked: the project was generated and verified with kicad-cli 7.0.11, but Homebrew offers only KiCad 10. Install 7.0.11 from kicad.org, KiCad 10 from Homebrew, or skip KiCad for now?

Options: 7.0.11, which matches the verified netlists (recommended at the time); KiCad 10, which is easier to keep updated, but its netlist export and libraries may differ; or no KiCad.

Chosen: KiCad 10 (10.0.7, installed by the owner).

Consequence: two things had to change for KiCad 10. `kicad-cli` 10 writes its netlist one token per line, so the parsers in `check_netlist.py` and `assign_passives.py` were changed to accept either layout. The results were identical to KiCad 7's, and `passive_fp.json` came out byte for byte the same. The generators copy library symbols into each sheet, so they are pointed at **KiCad 7.0.11's symbol library**, downloaded separately. With it, an unchanged sheet regenerates identically apart from UUIDs. KiCad 10's library would have changed every symbol on every regenerated sheet. That library choice was not put to the owner; it follows from this answer.

6b. Which ngspice

What happened: Homebrew's ngspice 47 failed three blocks that pass on the original machine. `injector_turnoff` and `boost_converter` never finished, and `metering_unit_pwm` missed by 3 %. The suite also stalled at random. Fixes for all three were found: a ramp on an on/off charger model, a step-size cap and a time-weighted average. But the random stall remained.

The owner's position: first, that the circuits must work on the latest version too. Then, after the random stall: build ngspice 42 from source rather than spend more time on 47.

Chosen: ngspice 42, built from source into `~/opt/ngspice-42`. The committed suite passes 24/24 on it unchanged (25/25 with `power_path`).

Still open: the suite does not pass on ngspice 47 for the three causes above. Upgrading the simulator is a change that needs the whole suite re-run (`docs/HANDOFF.md` §2 and §5).

6c. Keep the ngspice 47 fixes or not

Asked: with 42 passing the committed code, revert the 47-specific edits to the netlists, `run_sim.py` and `ngrun.py`, and commit only the setup changes?

Chosen: revert. The 47 fixes are not in the repository; §6b records what they were.

7. The overvoltage cut-off in front of the sensor trackers (8 October)

§4 chose a tracking regulator per sensor group "behind a 100 V-class overvoltage cut-off", and named an LTC4368-class controller as the candidate. Its datasheet had never been retrieved. Retrieving it, and looking for an alternative, turned the candidate into a choice.

What the part is for

The three sensor regulators (TPS7B4253-Q1, one per group) survive at most **45 V** on their input. This board's battery rail reaches **73.3 V** for about 50 µs on ISO 7637-2 pulse 2a, a standard automotive surge test. So the regulators need a fast automatic switch in front of them:

- normally **on**, passing battery power through;
- above about 45 V it switches **off** within a few microseconds, and a 47 µF capacitor after it keeps the sensors powered meanwhile;
- when the surge has passed, it switches back **on**.

It is a small controller chip driving one power transistor. It must stay **on** through the 40 V load dump, which lasts 400 ms, far longer than the capacitor could bridge.

The question

Asked: which overvoltage cut-off controller should feed the three sensor trackers?

The first time it was asked, the options were given only as datasheet figures, and the owner replied that they could not decide between two part numbers. The question was then explained as above, with this

comparison:

	TPS48110-Q1 (Texas Instruments)	LTC4368 (Analog Devices)
Survives	100 V	100 V
Switches off in	2.6–4 µs after the threshold	under 8 µs
Automotive-qualified (AEC-Q100)	**yes**, Grade 1	**no**: C, I and H grades only (H = −40 to +125 °C)
Around it	a few more parts: charge-pump capacitor, unused current-sense pins tied off	simpler, 10 pins
Datasheet	SLUSEE5E (April 2026), from ti.com	Rev A, read through a mirror because analog.com timed out
Extra check needed	a 60 V/µs limit on how fast its input may rise	—

Recommended: TPS48110-Q1. Electrically, either would do the job. The deciding point is qualification. AEC-Q100 means the chip has passed the automotive industry's stress tests for temperature cycling, vibration and lifetime. Every other IC on this board has it, and the LTC4368 would have been the one exception, on the part that protects the sensor supply. The standing rule puts reliability before cost and convenience.

Chosen: TPS48110-Q1 ("ok use the recommended one").

Consequences:

- **Slew limit:** the input slew rate on pulse 2a's edge has to be checked against the 60 V/µs limit. The `sensor_rail` rewrite does that.
- **Threshold window:** the OV threshold must sit above the 40 V load dump at its lowest tolerance, and trip early enough that the regulators' input stays under 45 V. That includes the rise during the controller's 2.6–4 µs reaction time. The window is narrow, and the `sensor_rail` rewrite pins it.
- **The switch transistor** is the board's existing 100 V N-FET, IPD90N10S4-06, so there is one part number fewer to buy.

What the simulation then showed. Pulse 2a was applied through the real pin-21 filter. The trackers' input peaks at about **22 V**, nowhere near 45 V, at every corner. TI's own inrush network does most of that: the 22 nF from gate to ground holds the FET's gate, so the FET can only follow the surge up to its gate voltage. The OV comparator then opens it fully once the rail passes 40.5–42.1 V, and that timing turned out not to be critical. The 60 V/µs slew limit is met by feeding the controller's supply pins through 100 Ω and 22 nF, which gives 4.9 V/µs. The 47 µF hold-up is enough even aged to 33 µF with all three trackers at their limit. One thing it showed about the rest of the board: with the filter inductor not saturated, the clamp node reaches **76.6 V**, not the 73.3 V `transient_clamp.cir` reports. Every part there is 100 V class, and the figure is now a check.

A limit this design does not remove

The trackers' outputs survive −1 V to +45 V, so a sensor supply wire shorted to battery is survived at 24 V (jump start) and at 40 V (load dump). A short to battery **at the same time as** a pulse 2a surge (73.3 V) would exceed it. That is a double fault, and it is recorded rather than designed out.

8. The boost inductor, revisited (8 October)

§2b approved two changes to the injector boost converter: feed the controller's supply (VDD) from the clamped output side, and use a 100 µH inductor (Würth 7447709101). Building it showed the 100 µH part does not fit every rule.

What the inductor and its sense resistor have to do

The TPS40210-Q1 measures inductor current through a small sense resistor, and that one resistor sets two things:

- **Stability.** TI's Equation 9 caps the resistor at 80 % of VDD × L × fSW / (60 × (VOUT + VD − VIN)). Below it the converter is stable; above it, it goes "subharmonic" and oscillates at half its switching frequency. The worst corner is the 6 V cranking dip. Feeding VDD from the boost output raises VDD from about 5 V to about 32 V while cranking, which is the whole remedy.
- **The current limit.** The controller cuts each cycle when the resistor's voltage reaches 120–180 mV, a 1.5 : 1 spread over tolerance and temperature. The limit has to sit high enough never to trip in normal running (TI's Equation 49), and low enough that in a fault the inductor does not saturate badly.

The first question

Asked: with the approved 100 µH inductor, all three rules cannot be met strictly. Allow the fault current limit to reach the inductor's softer 30 % saturation point, or find a bigger inductor?

The owner's answer: "150 µH inductor comes in size 10x10x4 SMD package."

Checked against Würth's own datasheets:

- **7447714151, 10 × 10 × 4 mm:** 150 µH, Isat 1.4 A typ. It cannot work. Normal cranking peaks at about 1.14 A, so TI's rule puts the current limit's minimum near 1.8 A and its maximum near 2.7 A. The part would saturate in normal running, not just in a fault.
- **7447709151, 12 × 12 mm:** 150 µH, Isat 2.7 A (10 %) and 3.3 A (30 %) typ, in the **same footprint** as the approved 100 µH part.

The second question

Asked: which boost inductor?

	100 µH 7447709101 (as approved)	150 µH 7447709151
Sense resistor	50 mΩ	62 mΩ
Current limit, min–max	2.4–3.6 A	1.9–2.9 A
Limit max vs 10 % saturation	16 % over	7 % over
Limit max vs 30 % saturation	5 % under	14 % under
Stability margin (Eq. 9, 80 % rule)	1.07×	1.30×
No false trip while cranking (Eq. 49)	1.27×	1.07×

Recommended and chosen: 150 µH, Würth 7447709151, with a 62 mΩ sense resistor. It has more stability margin and less fault-current saturation, in the same footprint. This revises the 100 µH in §2b; the VDD-from-output remedy stands.

Why neither choice is strict: the 1.5 : 1 spread of the current- limit threshold means the limit's maximum is always 1.5 × its minimum. The minimum has to clear normal cranking current, so the maximum lands above the first softening point of any inductor this size. Partial saturation during a fault is not damaging: the controller still cuts the current every cycle.

How it was built

- **VDD from the output.** A follower off the 100 V rail gives VDD 34.3–45.0 V. It is 12V_GATE's circuit with a 39 V zener (BZG03C39, same Vishay family as the 36 V clamp): a PZTA42-Q pass transistor, 2 × 3.3 kΩ collector resistors, and a PMBT3904-Q limiting at 11.6 mA. A follower off a 100 V rail needs that protection whatever it feeds. If VDD shorted, the follower would otherwise burn.
- **Start-up from the battery.** The old 220 Ω feed became a 180 Ω start-up path, with its 36 V clamp, into a Schottky (SS20PH102, already on the board). The Schottky stops the follower pushing about 0.1 A back into the battery rail. 180 Ω, not 220 Ω, buys back the Schottky's drop: VDD is 4.77 V at the 6 V cranking dip, against the 4.5 V start threshold.
- **Found while building it:** the oscillator's timing resistor returns to VDD, and TI specifies the frequency −20 % / +7 % across VDD. Equation 9 scales with frequency, so the stability check has to use −20 %. At that corner the 150 µH design keeps 1.12×. The 100 µH design would have failed outright (42.9 mΩ against its 50 mΩ). The choice in this section was the right one for a reason nobody had seen yet.
- **Stated, not hidden:** the switch FET's datasheet gives only a typical gate charge. At twice that, start-up margin falls to 17 mV. Until the output passes about 36 V, the controller runs from the battery feed with a

shallow ramp, so subharmonic operation is possible for the first milliseconds of a cranking start, not in running.

Where these came from

Nothing here was found by reading the schematic. Each one surfaced when a part number was chosen and its datasheet was checked against what the drawing assumed — which is the same shape as every earlier finding in this project: a document and an executable check disagreeing, with nothing comparing them until someone did.